

Global Semiconductors and Semiconductor Capital Equipment

Global Semis: May 2026 WSTS Tracker - Sales 16.1% MoM, above typical (8.5% MoM) & +118.8% YoY on continued memory strength



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On Friday the SIA published its monthly WSTS report, which tracks sales, units and ASPs across the semiconductor industry, and provides details by products, end markets, and geographies. The recent report includes data up to and including May 2026.

Total semiconductor sales grew 118.8% YoY in May after rising 106.4% in April. YoY strength was again dominated by memory with sales more than quadrupling at 325.7% YoY, driving ~\$200B in incremental revenue YTD and with memory pricing alone driving ~2/3 of overall industry growth since the beginning of the year. Excluding memory, sales were still up strongly at 34.0% YoY.

Total semiconductor sales MoM were above typical seasonality, up 16.1% versus the historical May average of +8.5% MoM. On a MoM basis as-reported, product group performance was generally worse to in line with seasonal patterns.

DSP (3.3% vs. typical -2.5%) and NAND (40.7% vs. typical 26.0%) were better than typical MoM patterns. Discretes (-4.9% vs. typical 2.7%), Optoelectronics (-10.8% vs. typical -6.3%), Sensors & Actuators (-0.8% vs. typical 2.4%), Analog Std Linear (-5.4% vs. typical -3.5%), Analog App Specific (-9.3% vs. typical -0.6%), and DRAM (27.7% vs. typical 41.9%) were worse than typical MoM patterns. Logic (3.9% vs. typical 4.1%), MPU (5.0% vs. typical 5.2%), and MCU (-2.7% vs. typical -2.2%) were in line with typical MoM patterns.

Total unit shipments were down 2.7% MoM, with ASPs up 19.3% MoM.

By geography, total sales increased YoY and MoM in all regions.

Application specific ICs sales were up (2.4%) MoM, in line with typical historical patterns (up 3.0% MoM). 1 end markets was better than typical MoM, while 5 were worse than typical: Wired Communication (12.7% vs. typical 6.9%), was above typical MoM patterns. Consumer (-1.7% vs. typical 2.5%), Computer & Peripherals (3.8% vs. typical 5.7%), Wireless Communication (-3.2% vs. typical 1.5%), Automotive (-4.7% vs. typical 0.5%), and Multipurpose and Others (-5.2% vs. typical -4.0%) were worse than typical MoM patterns.

BERNSTEIN TICKER TABLE

Ticker	Rating	2 Jul 2026		Price Target	TTM Rel. Perf.	Reported EPS				Reported P/E (x)		
		Cur	Closing Price			Cur	2025A	2026E	2027E	2025A	2026E	2027E
AMD (Advanced Micro)	O	USD	517.82	600.00	256.3%	USD	4.17	6.98	14.61	124.1	74.2	35.4
ADI (Analog Devices)	M	USD	377.16	430.00	34.3%	USD	7.79	12.40	14.65	48.4	30.4	25.8
AVGO (Broadcom)	O	USD	360.45	550.00	11.8%	USD	6.82	11.60	18.69	52.8	31.1	19.3
INTC (Intel)	M	USD	120.35	100.00	416.0%	USD	0.43	1.07	1.50	282.3	112.4	80.0
NVDA (NVIDIA)	O	USD	194.83	315.00	3.1%	USD	4.77	9.19	12.52	40.8	21.2	15.6
NXPI (NXP Semiconductors)	M	USD	273.36	270.00	(1.4)%	USD	11.81	14.58	16.45	23.1	18.7	16.6
QCOM (Qualcomm)	M	USD	176.25	235.00	(10.5)%	USD	12.03	10.64	10.16	14.6	16.6	17.3
TXN (Texas Instruments)	M	USD	293.08	250.00	16.5%	USD	5.45	7.62	8.22	53.8	38.5	35.6
KLAC	O	USD	235.55	197.50	135.6%	USD	3.33	3.69	5.12	70.8	63.8	46.0
LRCX (Lam Research)	O	USD	351.41	340.00	236.5%	USD	4.14	5.68	7.98	85.0	61.9	44.1
AMAT (Applied Materials)	O	USD	603.04	525.00	196.5%	USD	9.42	12.17	15.56	64.0	49.6	38.8
TSM (TSMC)	O	USD	434.16	430.00	65.7%	USD	10.48	16.07	19.70	13.1	9.7	7.4
2330.TT (TSMC)	O	TWD	2,445.00	2,780.00	87.5%	TWD	66.25	101.62	124.63	11.7	8.6	6.6
2303.TT (UMC)	U	TWD	170.50	47.00	251.9%	TWD	3.34	3.31	3.85	5.6	5.5	5.2
5347.TT (Vanguard)	M	TWD	193.50	94.00	68.0%	TWD	4.30	4.34	4.44	5.7	5.7	5.7
2454.TT (MediaTek)	O	TWD	4,195.00	4,380.00	187.4%	TWD	66.17	69.52	170.50	63.4	60.3	24.6
3034.TT (Novatek)	M	TWD	538.00	370.00	(38.6)%	TWD	26.87	24.21	25.10	20.0	22.2	21.4
005930.KS (SEC- Samsung)	O	KRW	314,500	440,000	359.0%	KRW	6,611.53	48,393	77,273	47.6	6.5	4.1
005935.KS (SEC-Pref - Samsung)	O	KRW	208,000	374,000	256.9%	KRW	6,611.53	48,393	77,273	31.5	4.3	2.7
SMSN.LI (Samsung)	O	USD	5,040.00	7,350.00	298.8%	USD	116.15	812.39	1,290.80	43.4	6.2	3.9
000660.KS (SK hynix)	O	KRW	2,424,000	3,300,000	758.3%	KRW	60,341	395,677	568,862	40.2	6.1	4.3
MU (Micron)	O	USD	975.56	1,300.00	678.6%	USD	8.29	67.39	158.99	117.7	14.5	6.1
285A.JP (KIOXIA)	U	JPY	83,300	40,000	3362.2%	JPY	1,014.00	10,013	9,656.84	82.1	8.3	8.6
688012.CH (AMEC)	O	CNY	413.69	500.00	202.8%	CNY	3.40	4.95	7.18	121.7	83.6	57.6
688041.CH (Hygon)	O	CNY	325.40	450.00	102.2%	CNY	1.10	2.04	3.59	295.8	159.3	90.6
002371.CH (NAURA)	O	CNY	816.00	680.00	105.6%	CNY	5.66	10.22	16.41	144.1	79.9	49.7
688072.CH (Piotech)	O	CNY	832.00	580.00	388.1%	CNY	3.32	8.12	12.40	250.6	102.4	67.1
6415.TT (Silergy)	O	TWD	605.00	600.00	34.3%	TWD	6.40	9.81	15.50	94.5	61.7	39.0
ASML.NA (ASML)	O	EUR	1,634.40	2,300.00	120.2%	EUR	24.72	32.73	48.68	66.1	49.9	33.6
ASML (ASML)	O	USD	1,769.32	2,623.00	103.5%	USD	27.95	37.01	55.05	55.4	41.8	28.1
6857.JP (Advantest)	O	JPY	29,345	39,200	128.8%	JPY	514.52	735.65	870.09	57.0	39.9	33.7
6146.JP (DISCO)	O	JPY	76,590	85,000	42.8%	JPY	1,246.28	1,733.62	2,127.33	61.5	44.2	36.0
7741.JP (Hoya)	O	JPY	25,930	33,100	9.8%	JPY	744.42	802.89	946.43	34.8	32.3	27.4
4062.JP (Ibiden)	O	JPY	23,345	23,900	589.2%	JPY	214.91	259.64	411.19	108.6	89.9	56.8
6525.JP (Kokusai)	O	JPY	12,000	8,240.00	208.5%	JPY	128.63	200.23	260.07	93.3	59.9	46.1
6920.JP (Lasertec)	O	JPY	47,650	50,000	109.3%	JPY	937.82	893.18	976.61	50.8	53.3	48.8
6723.JP (Renesas)	O	JPY	4,820.00	6,300.00	108.3%	JPY	181.61	244.76	298.54	26.5	19.7	16.1
3436.JP (SUMCO)	M	JPY	5,094.00	3,190.00	309.3%	JPY	(33.61)	(57.32)	47.60	(151.6)	(88.9)	107.0
8035.JP (Tokyo Electron)	O	JPY	73,200	59,200	123.2%	JPY	1,250.88	1,504.14	1,848.77	58.5	48.7	39.6
SOI.FP (Soitec)	O	EUR	117.30	180.00	118.7%	EUR	(6.17)	0.28	2.49	(19.0)	425.5	47.0
MELE.BB (Melexis)	O	EUR	78.55	72.00	(15.2)%	EUR	2.78	2.87	3.74	28.2	27.4	21.0
XFAB.FP (X-Fab)	O	EUR	7.86	12.80	(4.8)%	USD	0.23	0.07	0.51	22.0	57.3	17.4
7735.JP (Screen)	M	JPY	18,940	12,600	168.4%	JPY	486.61	572.60	662.24	38.9	33.1	28.6
SPX			7,483.24									
ASIAX			1,977.84									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

AMD, ADI, AVGO, INTC, NVDA, NXPI, QCOM, KLAC, LRCX, AMAT, MU, 6723.JP, SOI.FP estimate is Adjusted EPS; AMD, ADI, AVGO, INTC, NVDA, NXPI, QCOM, KLAC, LRCX, AMAT, MU, 6723.JP, SOI.FP valuation is Adjusted P/E (x); TSM, 2330.TT, 2303.TT, 5347.TT valuation is P/B (x); XFAB.FP valuation is EV/EBIT (x); NVDA, SOI.FP base year is 2026;

Source: Bloomberg, Bernstein estimates and analysis.

Ticker	Rating	2 Jul 2026		Price Target	TTM Rel. Perf.	Reported EPS				Reported P/E (x)		
		Cur	Closing Price			Cur	2025A	2026E	2027E	2025A	2026E	2027E
EM			1,835.33									
EDME			1,622.79									
JPL			2,655.16									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

AMD, ADI, AVGO, INTC, NVDA, NXPI, QCOM, KLAC, LRCX, AMAT, MU, 6723.JP, SOI.FP estimate is Adjusted EPS; AMD, ADI, AVGO, INTC, NVDA, NXPI, QCOM, KLAC, LRCX, AMAT, MU, 6723.JP, SOI.FP valuation is Adjusted P/E (x); TSM, 2330.TT, 2303.TT, 5347.TT valuation is P/B (x); XFAB.FP valuation is EV/EBIT (x); NVDA, SOI.FP base year is 2026;

Source: Bloomberg, Bernstein estimates and analysis.

INVESTMENT IMPLICATIONS

ADI (Market-Perform, \$430.00): ADI is executing well with both cyclical and idiosyncratic drivers, though may be closer to peak, and valuations remain elevated.

AMD (Outperform, \$600.00): Expectations remain high, but exposure to AI demand driving both a CPU and GPU story can provide substantial growth.

AVGO (Outperform, 550.00): A strong 2026 AI trajectory seems set to accelerate into 2027 and beyond, bolstered by software, cash deployment, and superb margins & FCF.

INTC (Market-Perform, \$100.00): Server strength is helping the company get back on their feet, and narrative/headlines may fuel the vibe for now.

NVDA (Outperform, \$315.00): The datacenter opportunity is enormous, and still early, with material upside still possible.

NXPI (Market-Perform, \$270.00): The pace and makeup of recovery remains open for debate.

QCOM (Market-Perform, \$235.00): Memory headwinds appear likely to pressure smartphone builds and numbers still appear high though the datacenter narrative sounds promising.

TXN (Market-Perform, \$250.00): TXN shares feel fully valued in the current environment.

AMAT (Outperform, \$525.00): We maintain a positive view on secular WFE growth and see a number of drivers for AMAT including SAM growth, an increasing services narrative, and capital return.

LRCX (Outperform, \$340.00): The company is benefiting from key inflections (GAA, packaging, HBM, NAND upgrades) and CY26 commentary seems supportive.

KLAC (Outperform, \$197.50): Amid positive WFE trends KLAC possesses structural growth drivers, a strong and durable competitive position, lower China replacement risk, and disciplined capital allocation, warranting premium valuation.

TSMC: We rate TSMC Outperform with PT=NT\$2,780.00.

UMC: We rate UMC Underperform with PT=NT\$47.00.

Vanguard: We rate Vanguard Market-Perform with PT=NT\$94.00.

Novatek: We rate Novatek Market-Perform with PT=NT\$370.00.

MediaTek: We rate MediaTek Outperform with PT=NT\$4,380.00.

Samsung Electronics: We rate Samsung Electronics Outperform with target price of KRW 440,000.

SK hynix: We rate SK hynix Outperform with target price to KRW 3,300,000.

Micron: We rate Micron Outperform with target price of US\$1,300.00.

KIOXIA: We rate KIOXIA Underperform with target price of JPY40,000.

MELE.BB: We rate Melexis Outperform with target price of €72.00.

SOI.FP: We rate Soitec Outperform with target price of €180.00.

XFAB.FP: We rate X-FAB Outperform with target price of €12.80

NAURA: We rate Naura Outperform with PT of CNY 680.00.

AMEC: We rate AMEC Outperform with PT of CNY 500.00.

Piotech: We rate Piotech Outperform with PT of CNY 580.00.

Hygon: We rate Hygon Outperform with PT of CNY 450.00.

Silergy: We rate Silergy Outperform with PT of TWD 600.00.

DISCO: We rate DISCO Outperform with PT of JPY 85,000.

Tokyo Electron: We rate Tokyo Electron Outperform with PT of JPY 59,200.

ASML: We rate ASML Outperform with PT of EUR 2,300.00.

Advantest: We rate Advantest Outperform with PT of JPY 39,200.

Screen: We rate Screen Market-Perform with PT of JPY 12,600.

Kokusai: We rate Kokusai Outperform with PT of JPY 8,240.00.

Lasertec: We rate Lasertec Outperform with PT of JPY 50,000.

SUMCO: We rate SUMCO Market-Perform with PT of JPY 3,190.00.

Hoya: We rate Hoya Outperform with PT of JPY 33,100.

Ibiden: We rate Ibiden Outperform with PT of JPY 23,900.

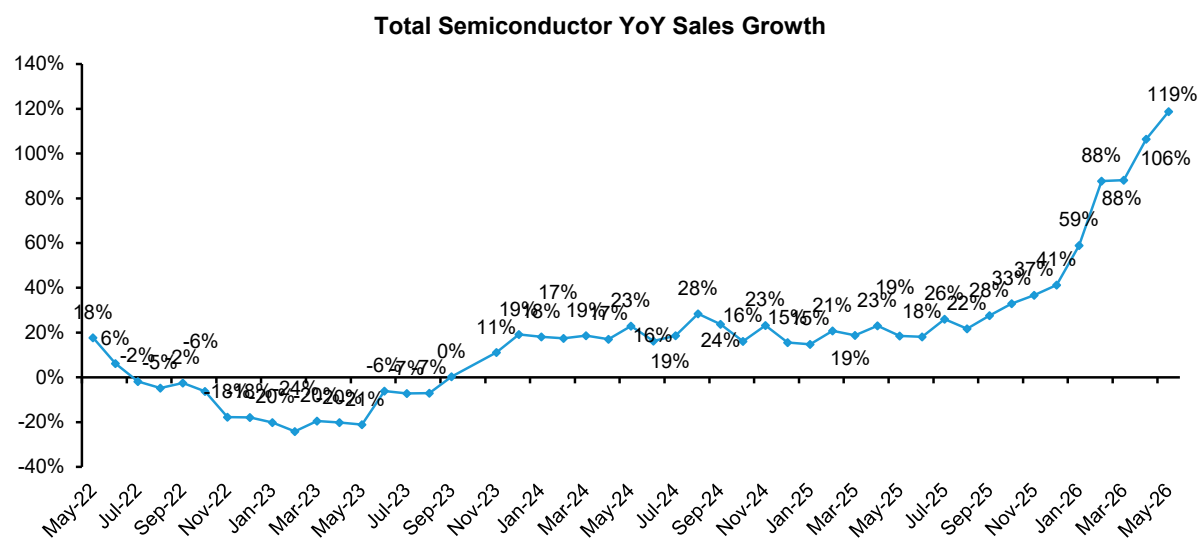
Renesas: We rate Renesas Outperform with PT of JPY 6,300.00.

DETAILS

On Friday the SIA published its monthly WSTS report, which tracks sales, units and ASPs across the semiconductor industry, and provides details by products, end markets, and geographies. The recent report includes data up to and including May 2026.

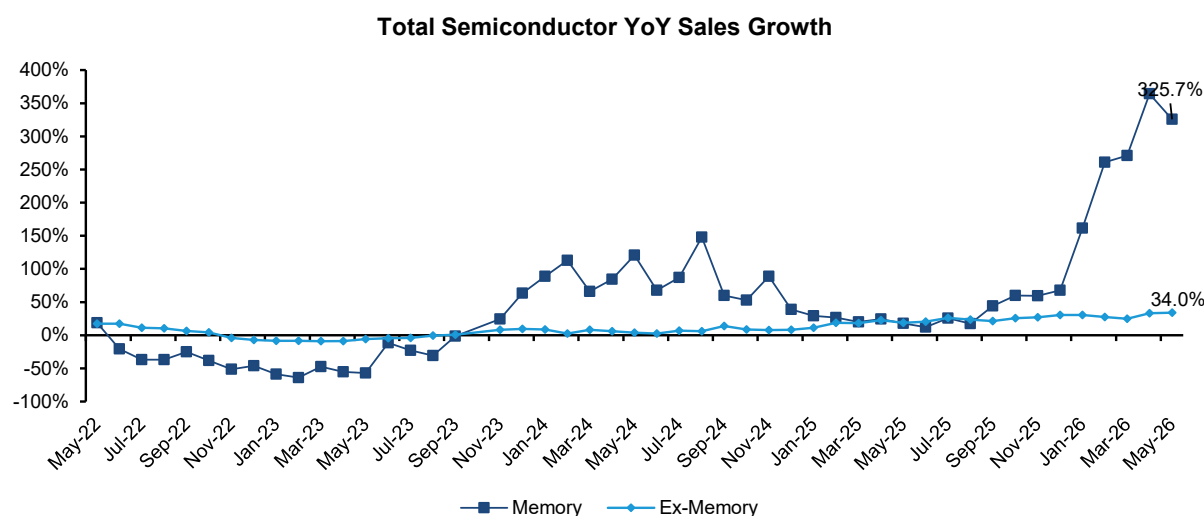
Total semiconductor sales grew 118.8% YoY in May after rising 106.4% in April (**Exhibit 1**). YoY strength was again dominated by memory with sales more than quadrupling at 325.7% YoY, driving ~\$200B in incremental revenue YTD and with memory pricing alone driving ~2/3 of overall industry growth since the beginning of the year (**Exhibit 3**, **Exhibit 4**). Excluding memory, sales were still up strongly at 34.0% YoY (**Exhibit 2**). On a 3-month rolling basis sales were up 35.6%, much better than typical historical patterns of up MSD (5.0%) (**Exhibit 5**).

EXHIBIT 1: **Historical total semiconductor YoY sales growth**

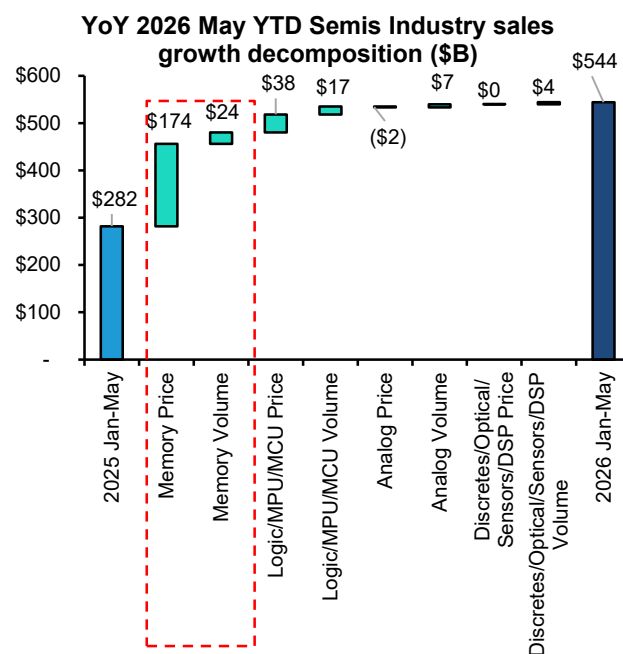


Source: WSTS, Bernstein Analysis

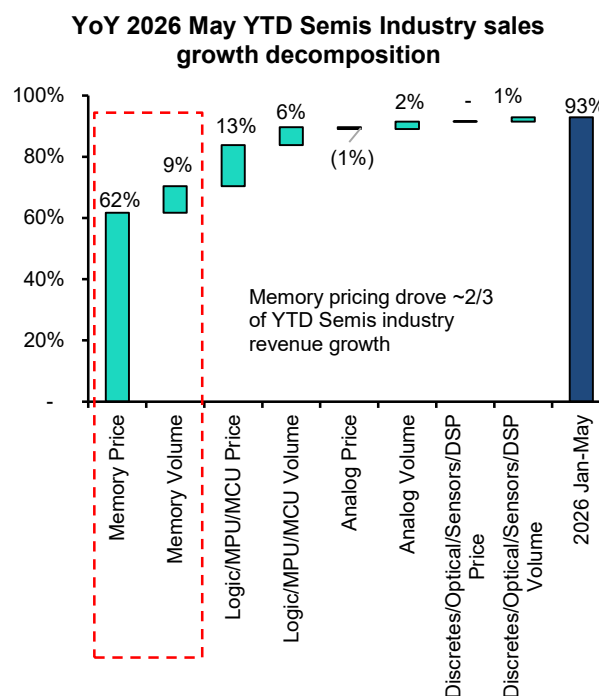
EXHIBIT 2: **Historical memory vs. non-memory YoY sales growth**



Source: WSTS, Bernstein Analysis

EXHIBIT 3: Memory drove ~\$200B in incremental Semis sales YTD through May...

Price x volume mix effects attributed to price.
Source: WSTS, Bernstein analysis

EXHIBIT 4: ...with Memory pricing alone driving ~2/3 of YTD Semis revenue growth

Price x volume mix effects attributed to price.
Source: WSTS, Bernstein analysis

EXHIBIT 5: Semiconductor industry % change in sales

	vs 1 month ago		vs 3 months ago		3 months rolling		vs 1 year ago
	This Month	Historical Average (20-25)	This Month	Historical Average (20-25)	This Month	Historical Average (20-25)	This month
Discretes	-4.9%	2.7%	11.5%	6.3%	11.8%	3.7%	14.8%
Optoelectronics	-10.8%	-6.3%	3.5%	-4.1%	1.2%	-3.5%	22.4%
Sensors & Actuators	-0.8%	2.4%	8.0%	5.5%	3.2%	2.1%	0.8%
Analog Std Linear	-5.4%	-3.5%	11.8%	9.2%	10.5%	5.8%	23.7%
Analog App Specific	-9.3%	-0.6%	1.2%	1.6%	-0.1%	0.1%	4.7%
Logic	3.9%	4.1%	13.5%	7.0%	10.7%	4.3%	48.5%
MPU	5.0%	5.2%	13.9%	4.0%	6.5%	0.4%	30.8%
MCU	-2.7%	-2.2%	18.3%	1.9%	8.7%	0.6%	19.6%
DSP	3.3%	-2.5%	35.6%	-2.0%	4.7%	1.6%	-1.5%
Memory	31.7%	32.7%	48.5%	24.2%	75.6%	14.6%	325.7%
DRAM	27.7%	41.9%	35.5%	24.9%	67.3%	16.1%	309.0%
NAND	40.7%	26.0%	80.2%	25.4%	95.9%	13.6%	377.1%
TOTAL	16.1%	8.5%	29.9%	8.9%	35.6%	5.0%	118.8%

Source: WSTS, Bernstein Analysis

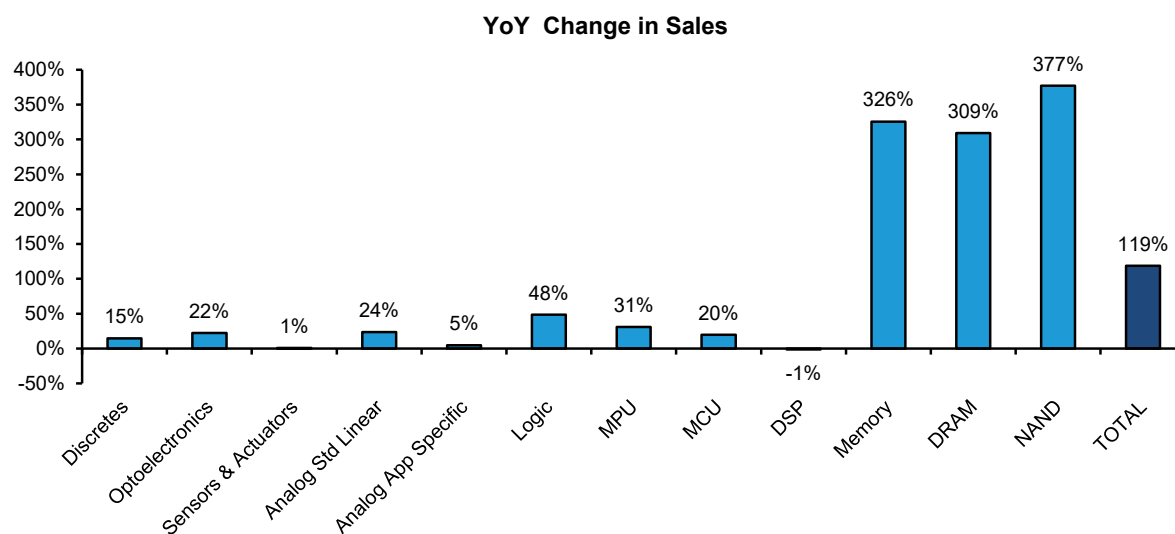
Total semiconductor sales MoM were above typical seasonality, up 16.1% versus the historical May average of +8.5% MoM. On a MoM basis as-reported, product group performance was generally worse to in line with seasonal patterns.

In May on a YoY basis and on a product group level, all product groups were up, except DSPs (which were down ~1% YoY) (**Exhibit 6**).

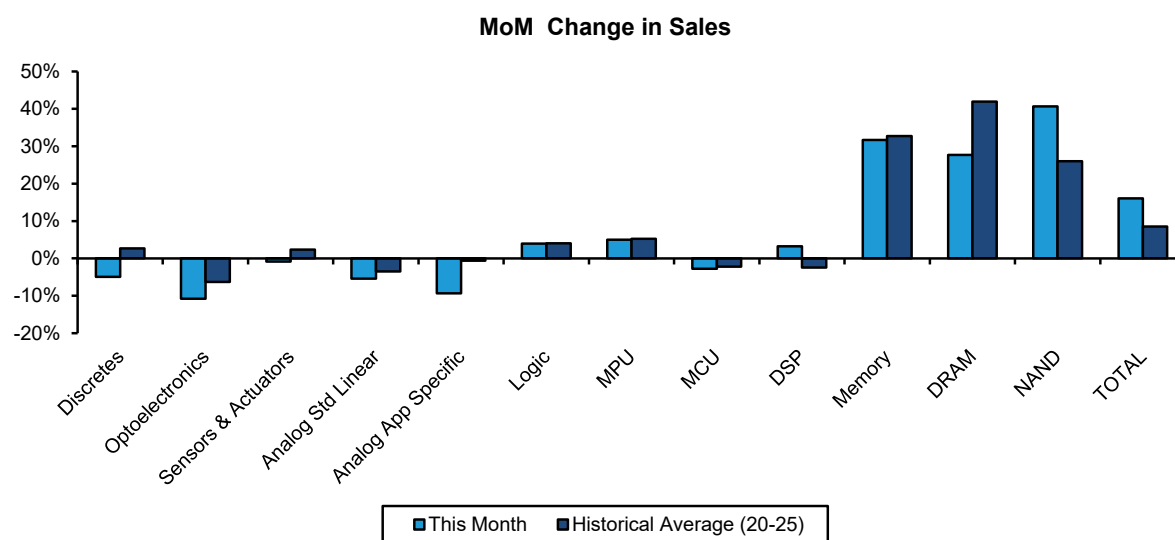
On a MoM basis, product group performance was generally worse to in line with typical seasonal patterns (**Exhibit 7**).

- DSP (3.3% vs. typical -2.5%) and NAND (40.7% vs. typical 26.0%) were better than typical MoM patterns.

- Discretes (-4.9% vs. typical 2.7%), Optoelectronics (-10.8% vs. typical -6.3%), Sensors & Actuators (-0.8% vs. typical 2.4%), Analog Std Linear (-5.4% vs. typical -3.5%), Analog App Specific (-9.3% vs. typical -0.6%), and DRAM (27.7% vs. typical 41.9%) were worse than typical MoM patterns.
- Logic (3.9% vs. typical 4.1%), MPU (5.0% vs. typical 5.2%), and MCU (-2.7% vs. typical -2.2%) were in line with typical MoM patterns.

EXHIBIT 6: **Total semiconductor YoY change in sales**

Source: WSTS, Bernstein Analysis

EXHIBIT 7: **Total semiconductor MoM change in sales**

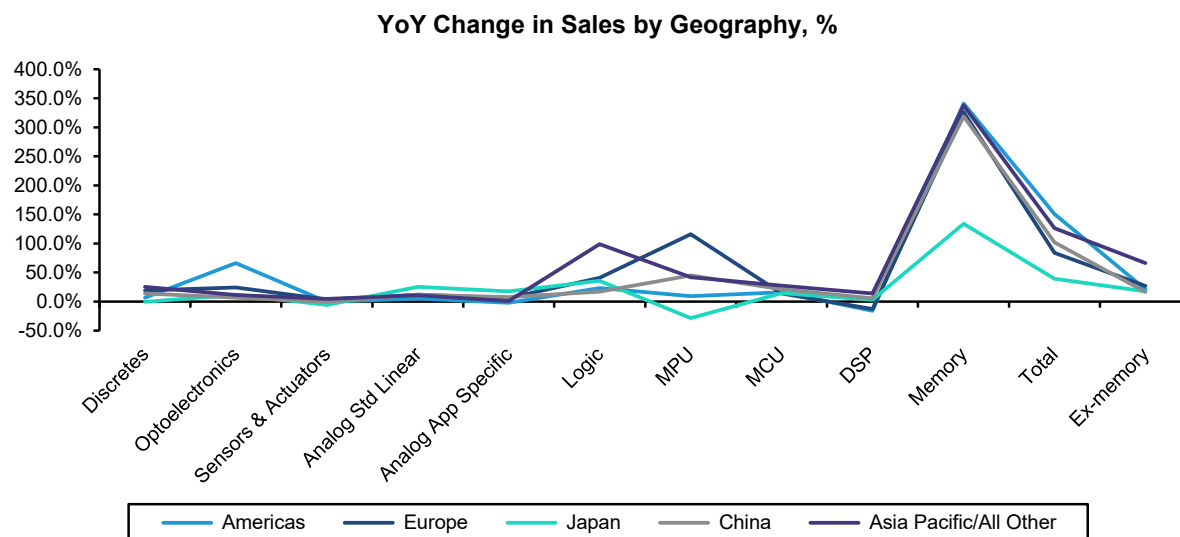
Source: WSTS, Bernstein Analysis

By geography, total sales increased YoY and MoM in all regions.

On a YoY basis, total semiconductors sales were up 150.7% in the Americas, 83.7% in Europe, 39.4% in Japan, 101.9% in China, and 126.3% in Asia Pacific/All Other (**Exhibit 8**).

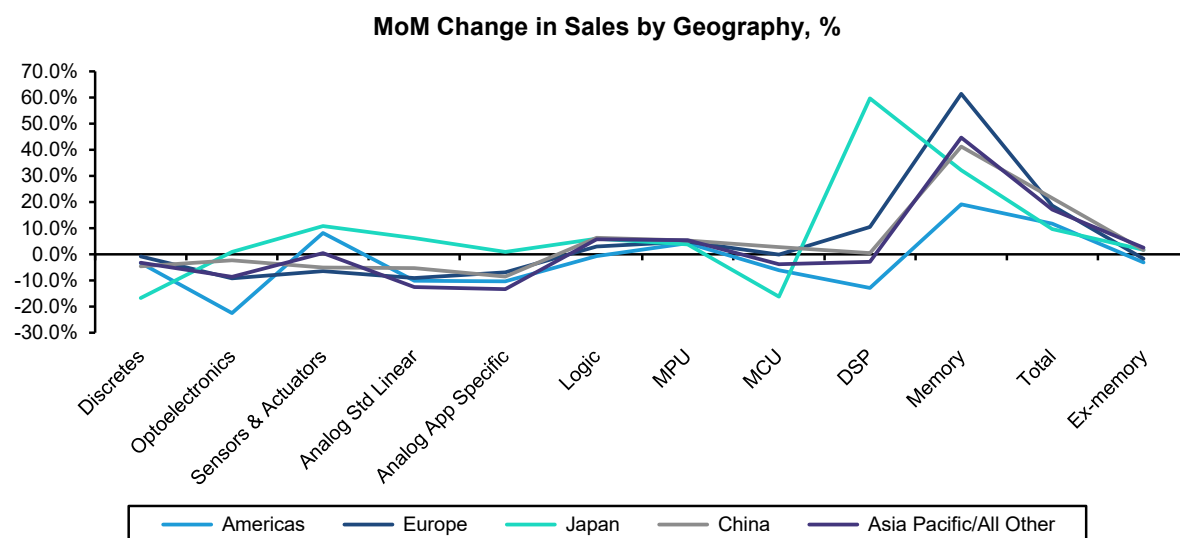
On a MoM basis, sales were up 11.8% in the Americas, 18.7% in Europe, 9.7% in Japan, 21.5% in China, and up 17.1% in Asia Pacific/All Other (Exhibit 9).

EXHIBIT 8: Total semiconductor YoY change in sales by geography



Source: WSTS, Bernstein Analysis

EXHIBIT 9: Total semiconductor MoM change in sales by geography



Source: WSTS, Bernstein Analysis

Total unit shipments were down 2.7% MoM, with ASPs up 19.3% MoM.

- Unit shipments were up MoM for 6 product groups, fell for 4, and were flat for 1.
- Units were up MoM for Sensors & Actuators, Logic, MPU, MCU, DRAM[bits], and NAND[bits], (2.2%, 1.8%, 3.6%, 1.8%, 13.2%, 19.5%, respectively)
- Units were down MoM for Discretes, Analog Std Linear, Analog App Specific and DSP (-5.8%, -2.3%, -6.4%, -8.0%, respectively)

- Units were flattish MoM for Optoelectronics (-0.5%).
- On a 3 months rolling basis overall shipments were up 9.1%.
- On a YoY basis shipments increased 12.7%. Shipments increased YoY for Discretes, Optoelectronics, Sensors & Actuators, Analog Std Linear, Analog App Specific, Logic, MPU, MCU, DRAM[bits], and NAND[bits], and fell for DSP. (**Exhibit 10**).

EXHIBIT 10: **Change in semiconductor unit shipments**% Change in Units

	vs 1 month ago	vs 3 months ago	3 months rolling	vs 1 year ago
Discretes	-5.8%	13.5%	8.5%	3.9%
Optoelectronics	-0.5%	3.5%	0.3%	17.2%
Sensors & Actuators	2.2%	11.3%	5.8%	8.6%
Analog Std Linear	-2.3%	18.5%	12.1%	24.3%
Analog App Specific	-6.4%	12.4%	14.3%	19.4%
Logic	1.8%	21.8%	13.9%	20.0%
MPU	3.6%	-2.6%	-6.5%	6.7%
MCU	1.8%	15.5%	4.3%	14.2%
DSP	-8.0%	32.3%	10.7%	-11.3%
DRAM (Based on Bit Shipped)	13.2%	-0.7%	8.5%	16.1%
NAND (Based on Bit Shipped)	19.5%	8.4%	6.5%	10.6%
TOTAL	-2.7%	13.8%	9.1%	12.7%

Source: WSTS, Bernstein Analysis

On a MoM basis, industry ASPs were up 19.3%, with 6 out of 11 product groups experiencing ASP increases and 5 groups seeing ASP declines.

- ASPs increased MoM for Discretes, Logic, MPU, DSP, DRAM[bits], and NAND[bits] (1.5%, 2.1%, 1.3%, 12.2%, 12.9%, 17.8%, respectively).
- ASPs fell MoM for Optoelectronics, Sensors & Actuators, Analog Std Linear, Analog App Specific, and MCU (-10.4%, -3.0%, -3.2%, -3.3%, -4.4%, respectively).
- Versus 3 months ago, ASPs rose in 5 out of 11 product groups, fell in 5 product groups and were flattish in 1. On a YoY basis ASPs rose in 8 out of 11 product groups, fell in 2 product groups and were flattish in 1. Total industry ASPs rose 94.2% YoY on the back of memory price increases. (**Exhibit 11**).

EXHIBIT 11: **Change in semiconductor ASPs**% Change in ASP

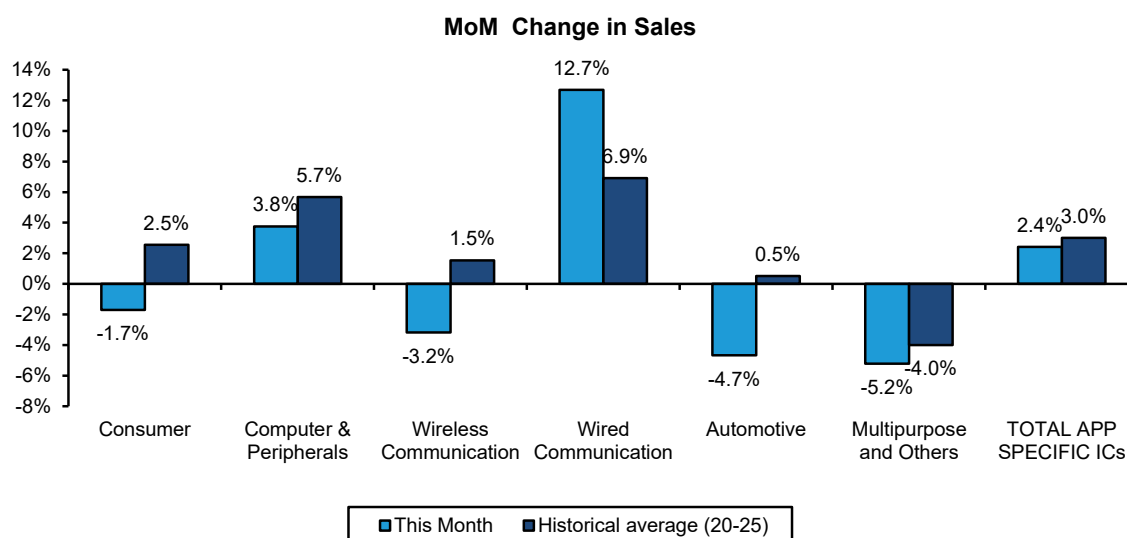
	vs 1 month ago	vs 3 months ago	vs 1 year ago
Discretes	1.5%	-1.5%	11.7%
Optoelectronics	-10.4%	0.0%	4.6%
Sensors & Actuators	-3.0%	-3.0%	-7.2%
Analog Std Linear	-3.2%	-5.5%	-0.4%
Analog App Specific	-3.3%	-10.0%	-12.4%
Logic	2.1%	-6.8%	23.7%
MPU	1.3%	16.9%	22.6%
MCU	-4.4%	2.4%	4.8%
DSP	12.2%	2.5%	11.1%
DRAM (Based on \$/Bit)	12.9%	36.5%	252.4%
NAND (Based on \$/Bit)	17.8%	66.3%	331.4%
TOTAL	19.3%	14.1%	94.2%

Source: WSTS, Bernstein Analysis

Application specific ICs sales were up (2.4%) MoM, in line with typical historical patterns (up 3.0% MoM). 1 end market was better than typical MoM, while 5 were worse than typical (Exhibit 12).

- Application specific ICs sales were up (2.4%) MoM, in line with typical historical patterns (3.0% MoM).

- 1 product group was above typical MoM, and 5 product groups were below typical MoM: Wired Communication (12.7% vs. typical 6.9%), was above typical MoM patterns.
- Consumer (-1.7% vs. typical 2.5%), Computer & Peripherals (3.8% vs. typical 5.7%), Wireless Communication (-3.2% vs. typical 1.5%), Automotive (-4.7% vs. typical 0.5%), Multipurpose and Others (-5.2% vs. typical -4.0%), were worse than typical MoM patterns.

EXHIBIT 12: **Change in sales by end markets (where WSTS data provides a split by application)**

Source: WSTS, Bernstein Analysis

By product category, we note the following:

Analog Standard Linear: Total Analog Standard Linear sales were down (-5.4%), worse than typical patterns (-3.5%). Shipments were down (-2.3%) with ASPs down (-3.2%). Product category changes MoM were as follows: Amplifiers/Comparators (-5.4% vs. typical -7.3%), Interface (-9.8% vs. typical +0.5%), Power Management (-3.6% vs. typical -2.1%), and Signal Conversion (-9.9% vs. typical -8.2%) (**Exhibit 13**).

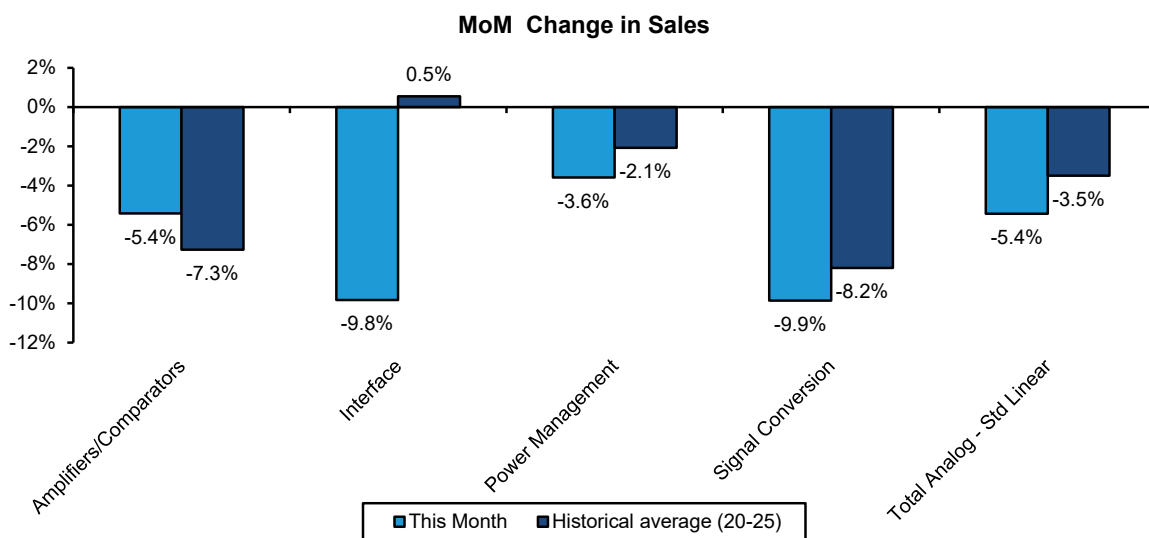
Analog Application Specific: Total analog application specific sales were down (-9.3%) MoM, worse than typical patterns (-0.6%). Units were down (-6.4%) MoM, with ASPs down (-3.3%). End markets changes MoM were as follows: Consumer (-8.1% vs. typical -1.0%), Computer (2.5% vs. typical 4.0%), Communications (-12.6% vs. typical -1.3%), Automotive (-5.3% vs. typical -1.4%), and Industrial & Others (-19.6% vs. typical of -6.9%) (**Exhibit 14**).

MPU: MPU sales were up 5.0% MoM, in line with typical patterns (5.2% MoM). Units were up (3.6%) MoM with ASPs up (1.3%) MoM. On a 3-months rolling basis (e.g. March-May vs. December-February), MPU sales were up (6.5%), better than typical average of 0.4%. Sales increased 13.9% vs. 3 months ago and increased 30.8% YoY.

MCU: MCU sales were down (-2.7%) MoM, in line with historical averages of down 2.2%, with units up (1.8%) and ASP down (-4.4%). Product category changes MoM were as follows: Consumer (-5.4% vs. typical -6.7%), Computer (-7.2% vs. typical 12.8%), Handsets (89.0% vs. typical -11.4%), Wireless Communication-All Other (-1.6% vs. typical -0.4%), Wired Communication (23.9% vs. typical 3.2%), Automotive (-5.9% vs. typical -0.1%), and Multi-purpose and others (-2.2% vs. typical 0.1%) (**Exhibit 15**).

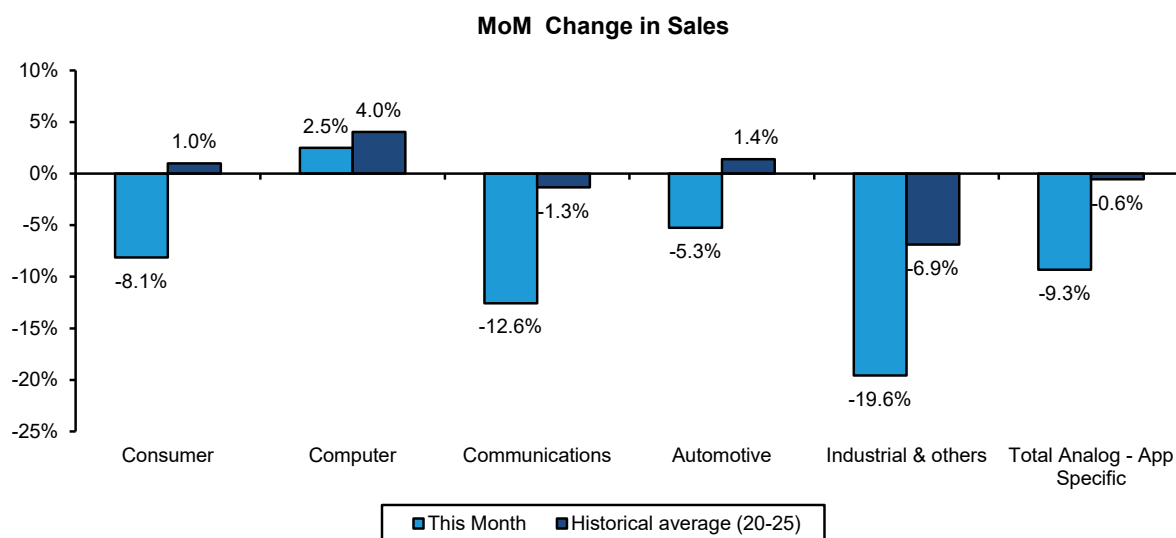
Memory: Memory sales were up (31.7%) MoM, in line with historical average of up 32.7%. On a bit shipped basis, DRAM shipments were up 13.2% MoM and NAND was up 19.5% MoM. On a \$ per bit basis, DRAM ASPs rose 12.9%, while NAND ASPs rose 17.8%. (**Exhibit 16, Exhibit 17**).

EXHIBIT 13: **Analog Standard Linear - MoM change in sales**



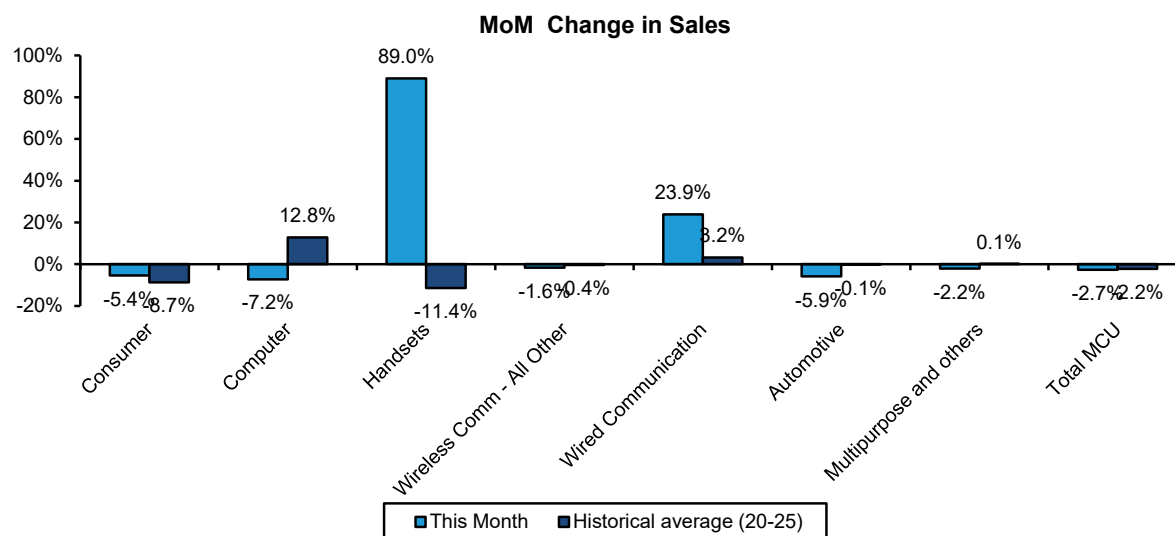
Source: WSTS, Bernstein Analysis

EXHIBIT 14: **Analog App Specific - MoM change in sales**



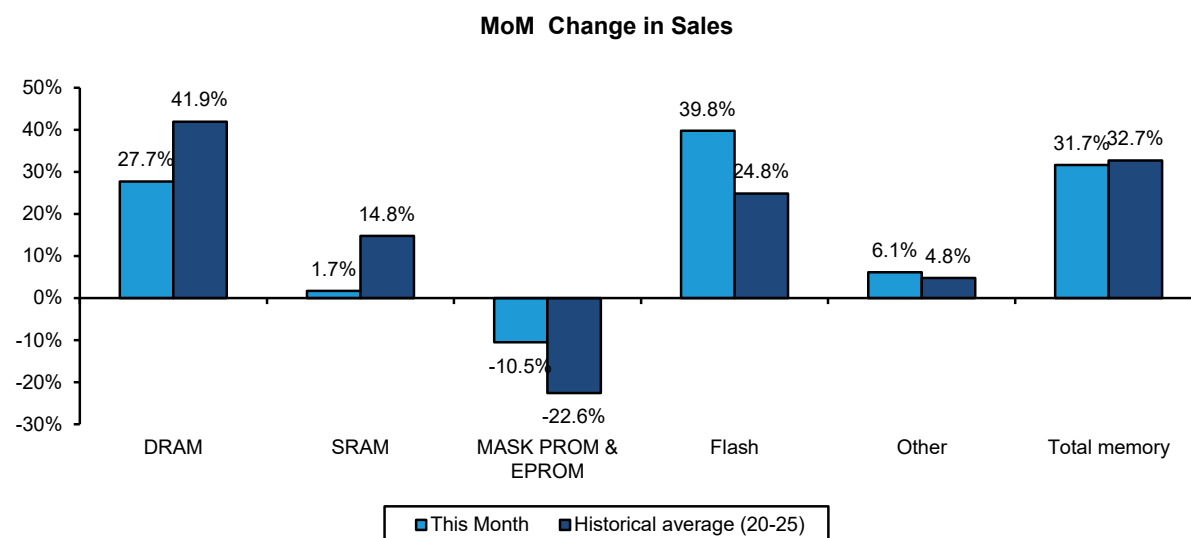
Source: WSTS, Bernstein Analysis

EXHIBIT 15: **MCU - MoM change in sales**



Source: WSTS, Bernstein Analysis

EXHIBIT 16: **Memory - MoM change in sales**



Source: WSTS, Bernstein Analysis

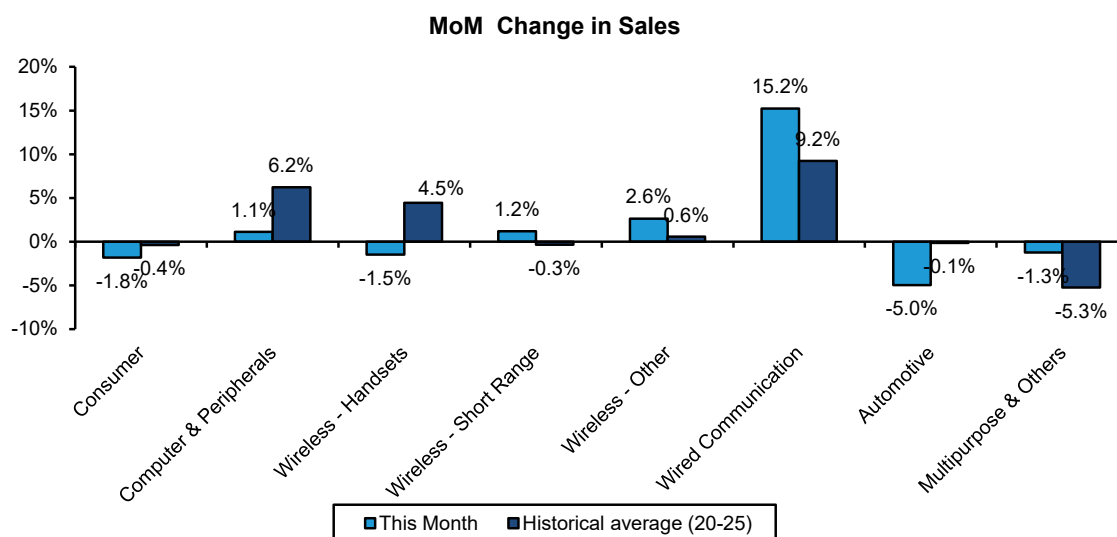
EXHIBIT 17: **Memory - Change in units and ASP on bit basis****% change based on bit shipped**

	vs 1 month ago	vs 3 months ago	3 months rolling	vs 1 year ago
DRAM	13.2%	-0.7%	8.5%	16.1%
NAND	19.5%	8.4%	6.5%	10.6%

% change in \$ per bit

	vs 1 month ago	vs 3 months ago	vs 1 year ago
DRAM	12.9%	36.5%	252.4%
NAND	17.8%	66.3%	331.4%

Source: WSTS, Bernstein Analysis

EXHIBIT 18: **MOS Special Purpose Logic - MoM change in sales**

Source: WSTS, Bernstein Analysis

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